

The Unified African Global Emissions Trading Architecture : A Fraud-Resistant Market Framework with Performance-Weighted Governance

(Working document March 2026)

Introduction

The Brazzaville Foundation could support through its partners in government and finance a new Unified African Global Emissions Trading Architecture for African countries that would be set as a framework to build within and through a resilient carbon reduction programme for all African countries supported by a clever democratic structure expected to be resilient to the challenges of the next one or two decade.

This would allow the continent to not only reduce carbon emission but also finance all sort of parts of the economy with the taxes levied on polluting industries and transfer to a more ecological economy with a simple, efficient and transparent market tool.

I – Context : the problem with carbon emission accounting and trading

Existing emissions trading systems suffer from three structural failures that incremental reform cannot resolve.

Narrow coverage.

The European Union Emissions Trading System, the world's largest carbon market, covers carbon dioxide and a limited set of industrial gases from approximately 10,000 installations. Methane from agriculture, black carbon from shipping, hydrofluorocarbons from refrigeration, mercury from artisanal mining, and reactive nitrogen from fertiliser production are either excluded or regulated through separate, uncoordinated instruments. The majority of environmental damage remains unpriced and many industries emitting carbon either also utilise the other pollutants too in their industrial processes or shift towards producing goods requiring them as ecological dumping under certain jurisdictions.

Vulnerability to manipulation.

Participants in current carbon markets self-report emissions, self-certify offsets, and submit bids whose truthfulness cannot be independently verified. Countries set their own nationally determined contributions with no binding enforcement. The system relies on voluntary compliance where the economic incentive to misreport is substantial and the probability of detection is low. Documented cases of fraudulent offset credits, inflated baselines, and strategic underreporting confirm this is not a theoretical concern.

Governance without accountability.

In the UNFCCC and related treaty bodies, every member state holds equal procedural standing regardless of its environmental record. A nation that has systematically failed to meet its commitments carries the same governance weight as one that has exceeded its targets. This produces a free-rider problem in which the least committed parties can block or dilute the ambition of the most committed.

This paper proposes a unified architecture—the Unified African Global Emissions Trading Architecture, or UAGETA—that addresses all three failures simultaneously through a system in which truthful participation is not merely encouraged but is the economically rational choice for every actor.

II – The Proposed System

A universal emissions permit

UAGETA introduces a single tradeable instrument—the Universal Emission Unit—that denominates permits for all regulated pollutants. Each substance is converted to this common unit through a damage-weighted equivalence factor reflecting its full social cost: its contribution to climate forcing, its atmospheric persistence, and its effects on human health.

- Carbon dioxide serves as the reference.
- A tonne of methane is equivalent to roughly 30 units;
- a tonne of sulphur hexafluoride to approximately 24,000, and henceforth - Table 1 below presents the conversions.

Table 1: UAGETA Pollutant Coverage and Equivalence Factors

Pollutant Class	Representative	DWEF (UEU/tonne)	Basis
Greenhouse Gases	CO ₂	1.00	Reference unit
Greenhouse Gases	CH ₄ (Methane)	28–36	GWP-100 + health
Greenhouse Gases	N ₂ O (Nitrous oxide)	265–298	GWP-100 + ozone
Fluorinated Gases	HFC-134a	1,300–1,530	GWP-100
Fluorinated Gases	SF ₆	23,500–25,200	GWP-100
Short-lived Pollutants	Black carbon	460–1,500	GTP-20 + health
Short-lived Pollutants	SO ₂	80–160	Health damage cost
Short-lived Pollutants	NO _x	40–120	Ozone + PM2.5 form.
Persistent Organics	PCBs, Dioxins	10,000–500,000	DALY-weighted toxicity
Heavy Metals	Mercury (Hg) etc	50,000–200,000	Neuro-toxicity / DALY

Nutrient Loading	Reactive nitrogen	15–45	Eutrophication damage
Plastics	Microplastics	Coverable under study	Marine/health damage

A cheat-proof auction

Permits are allocated through a sealed-bid auction whose payment rule is designed so that every participant's profit-maximising strategy—regardless of what anyone else does—is to bid its true willingness to pay.

This is a structural property of the auction design, not a behavioural assumption. The mechanism charges each participant the cost that its participation imposes on all other participants, that is, **the polluter is charged the cost of all regional environmental externalities**. A firm that overstates its demand pays more than it gains. A firm that understates its demand forfeits permits it valued more than it saved. The only profitable action is honesty. This means that this property holds independently for every participant. A government cannot benefit its domestic firms by instructing them to bid strategically; the firms themselves would refuse because misreporting reduces their own surplus. The auction is self-financing: total payments cover total costs with no external subsidy required.

Step 1 : Allocation rule, everyone reports their valuation, and permits go to whoever values them most.

Step 2 : Payment rule, this is where the. Each firm pays not its own bid but the **harm its presence causes to everyone else**. Concretely, the system calculates what the total surplus of all other firms would have been if firm *i* didn't exist, then subtracts what those other firms actually get with firm *i* present. The difference is firm *i*'s payment.

Here is a concrete example.

Suppose there's one permit and three firms. Firm A values it at €90, Firm B at €70, Firm C at €30. The permit goes to Firm A (highest value). What does A pay? Without A, the permit would go to B, giving B a surplus of €70. With A present, B gets nothing — surplus of €0. So A's presence cost B exactly €70 of surplus. A pays €70. Notice that A valued the permit at €90, got it, and paid €70. A's profit is €20. Now ask — could A have done better by lying?

If A reports €60 instead of €90, A still wins ($€60 > €70$ is false — actually A would lose the permit). So understating is dangerous: you might lose something you wanted. If A reports €150 instead of €90, A still wins and still pays €70 (because the payment depends on others' bids, not A's). Overstating changes nothing about the payment but risks winning items you didn't actually want.

The only safe, optimal strategy is to report your true value. This holds regardless of what B and C do, regardless of whether they're colluding, regardless of whether A even knows they exist. Truth-telling is what game theorists call a **dominant strategy**.

At the level of all governments and companies trading carbon credits, this system is actually securing fairness and efficiency as well as trust and environmental security and international cooperation. It is setting them to be accountable as well as protected environmentall by all the other people contributing to the carbon market.

Table 2: Manipulation Channels and Formal Countermeasures

Cheating Channel	How It Works
Valuation fraud	Misreport abatement costs to obtain cheap permits
Emission falsification	Under-report actual emissions
Offset fabrication	Claim false carbon removals
Vote manipulation	Use economic/diplomatic leverage on governance
Leakage gaming	Relocate emissions to unmonitored jurisdictions
Price manipulation	Corner the permit market or collude

III – The Detecting and Deterring Falsfiction

Even with a truthful auction, countries may attempt to falsify their underlying emissions data. UAGETA addresses this through a three-layer verification architecture.

Market-based detection.

If a country's reported emissions are lower than its actual emissions, its firms will need more permits than their reports suggest. This need creates a detectable statistical signature in their trading behaviour: an abnormal sensitivity of the permit price to order flow from that country's firms. UAGETA operates a continuous monitor estimating this sensitivity for each country in real time. When it deviates significantly from what reported emissions would predict, the system flags a probable discrepancy.

Satellite verification.

Atmospheric concentrations of key pollutants are independently observable from space. The Verification Bureau cross-references national self-reports against satellite-derived estimates. Where the two diverge beyond a defined tolerance, the country must explain the discrepancy or accept a corrected figure.

On-site audit.

Countries triggering alerts from either layer are subject to inspection by internationally composed audit teams. No country may refuse without incurring an automatic governance penalty. Confirmed falsification triggers a penalty of three times the underpaid permit cost,

a reduction in governance weight, mandatory quarterly audits for three years, and—for repeat offenders within a decade—suspension of voting rights. These consequences are treaty-embedded and activate by formula, not political discretion.

IV – Performance-weighted governance

The most consequential innovation in UAGETA is that each country's influence over collective decisions is a continuous function of its independently verified environmental performance.

Each member state receives a Composite Neutrality Score between zero and one calculated from real statistics of environmental damage, computed from five components: the rate at which it is reducing emissions relative to its binding trajectory; its cumulative compliance with its neutrality pathway; its willingness to submit to verification; its financial contributions to global climate adaptation; and the inverse of its historical falsification record. No single component can be gamed in isolation because the score requires strong performance across all five dimensions, and each is verified from at least three independent data sources—self-reported data, satellite observation, and third-party audit—fused through a statistical methodology that converges to the true value as evidence accumulates.

The **Composite Neutrality Score (CNS)** computed as:

$$CNS_i(t) = \alpha_1 R_i(t) + \alpha_2 T_i(t) + \alpha_3 V_i(t) + \alpha_4 C_i(t) + \alpha_5 F_i(t)$$

where each component is normalised to [0, 1] and the weights $\alpha_1 + \dots + \alpha_5 = 1$ are set by treaty:

Component	Symbol	Weight	What It Measures	Verification Method
Reduction Rate	$R_i(t)$	0.30	Year-on-year UEU emission reduction relative to binding trajectory	Satellite + MRV + CEIM
Trajectory Compliance	$T_i(t)$	0.25	Cumulative deviation from pledged neutrality pathway	Independent technical panel
Verification Openness	$V_i(t)$	0.15	Willingness to submit to on-site audits and data transparency	Audit acceptance rate
Contribution Equity	$C_i(t)$	0.15	Financial contributions to climate adaptation and loss & damage	Fund disbursement records
Falsification Record	$F_i(t)$	0.15	Inverse of historical manipulation penalties	CEIM audit outcomes

The mapping from score to voting weight is deliberately convex: voting weight is proportional to the **square** of the score. A country scoring 0.8 holds four times the influence of a country scoring 0.4. This creates a powerful incentive gradient in which marginal improvements in environmental performance yield increasing marginal returns

in institutional power. Countries near the top compete for climate leadership because each additional point of performance translates into a disproportionate gain in governance authority. Conversely, poor performers are severely penalised: a country scoring 0.2 holds one-sixteenth the weight of a country scoring 0.8. A coalition of underperformers attempting to block ambitious policy would need an implausibly large membership to reach the one-third weighted threshold required to obstruct fundamental decisions.

To prevent strategic timing, the score is computed as a five-year exponentially weighted average. A country cannot inflate its score through a temporary performance spike before a critical vote. Sustained improvement over years is required to shift governance weight materially.

Decision-Making Thresholds

Different decisions require different supermajority thresholds under neutrality-weighted voting:

Table 3 : Decision-Making Thresholds

Decision Type	Weighted Vote Threshold	Rationale
Routine operational (auction schedules, reporting deadlines)	Simple majority (>50%)	Low stakes, efficiency
Cap adjustments, DWEF updates	Qualified majority (>60%)	Material but revisable
Admission of new pollutant classes	Supermajority (>67%)	Significant system expansion
Treaty amendments (governance formula, penalties)	Constitutional (>75%)	Foundational; protect against capture
Suspension of member voting rights	Constitutional (>75%) excl. target	Severest sanction

The seventy-five per cent constitutional threshold ensures that the governance formula itself cannot be weakened without overwhelming consensus—consensus that, by the structure of the weighting, underperformers cannot provide. Countries progressing toward carbon neutrality receive an enhanced governance bonus, creating a direct institutional reward for exceeding, not merely meeting, environmental commitments. The pathway from candidate status through provisional membership, full membership, and eventual climate leader or carbon-neutral status is transparent, rules-based, and determined entirely by the score formula—not by political negotiation.

The Neutrality Pathway: Evaluation Timeline

A country's journey through UAGETA is structured as a continuous evaluation of its neutrality commitment:

Table 4: Neutrality Pathway Stages

Stage	Duration	CNS Range	Governance Status	Key Obligations
Candidate	1–2 years	Not yet scored	Observer (no vote)	Submit baseline inventory, ratify treaty
Provisional Member	3–5 years	0.00–0.30	Reduced weight (×0.5)	First reduction targets, quarterly MRV
Full Member	Ongoing	0.30–0.70	Full weight per CNS	Binding annual targets, annual audits
Climate Leader	Ongoing	0.70–0.90	Enhanced weight (×1.2 bonus)	Exceeded targets, mentor developing nations
Carbon Neutral	Upon certification	0.90–1.00	Maximum weight (×1.5 bonus)	Net-zero across all UEU classes, verified

The critical innovation is that progress through these stages is determined entirely by the CNS formula and its independent verification—not by political negotiation. A country cannot claim carbon neutrality and receive the associated governance bonus unless its CNS score, computed from independently verified data using the Bayesian fusion methodology of Theorem 10.3, exceeds 0.90 sustained over at least 3 years.

V - Border Adjustments and International Fairness

UAGETA applies a border carbon adjustment to goods imported from countries with lower effective carbon prices, calculated as the difference in effective carbon prices multiplied by the product's carbon intensity. The effective price includes not only explicit carbon taxes and permit prices but also implicit pricing through fuel duties, efficiency mandates, and renewable energy subsidies. Any country that raises its carbon price to the UAGETA level sees its adjustment reduced to zero, creating a direct economic incentive for convergence without diplomatic coercion. The adjustment is formula-driven, publicly auditable, and administered by a dedicated office with no discretionary authority.

VI – Implementation Pathway

Phase 1 (Years 1–3).

A founding coalition—likely centred on the African Union and allied jurisdictions—establishes the institutional framework, appoints the Scientific Advisory Panel, and conducts pilot multi-pollutant auctions covering carbon dioxide and methane. Satellite verification infrastructure is deployed. Existing permits in the African Union and regional carbon trading systems are recognised at their equivalence factor values.

Phase 2 (Years 4–7).

Membership opens to all nations. Performance-weighted governance activates with a moderately convex weighting that tightens over time. The border adjustment begins operation. Coverage expands to nitrous oxide, hydrofluorocarbons, black carbon, and sulphur dioxide. Real-time market-based emissions monitoring begins.

Phase 3 (Years 8–12).

Governance convexity reaches its full level. All pollutant classes are incorporated. Derivative markets—options, futures, and swaps on Universal Emission Units—become fully operational. A tipping-adjusted price floor activates, ensuring permit prices automatically rise as atmospheric concentrations approach scientifically identified danger thresholds, without requiring politically contentious cap renegotiations.

VII - Rationale

The design of UAGETA rests on a simple observation: existing systems fail not because countries are unwilling to cooperate but because the institutional structure makes defection rational. A system that relies on voluntary compliance in the presence of strong incentives to defect will produce defection. A system that makes compliance the individually rational choice—regardless of what others do—will produce compliance.

The auction makes truthful bidding the profit-maximising strategy, removing the temptation to manipulate valuations. The verification architecture makes falsification statistically detectable and financially punitive, removing the temptation to underreport. The governance weights make genuine environmental improvement the path to institutional influence, removing the temptation to obstruct. The border adjustment makes relocation to weaker jurisdictions unprofitable, removing the temptation to leak.

None of these properties depends on trust, goodwill, or shared values. Each is a structural feature that holds under the assumption of fully self-interested behaviour. The system does not ask countries to be virtuous. It creates conditions under which self-interest and the global interest coincide.

The formal mathematical foundations of every claimed property—truthfulness of the auction, detectability of falsification, robustness of the governance weights to manipulation, stability of the border adjustment equilibrium, and accuracy of the pricing model under climate tipping point uncertainty—are established in the companion research paper from which this proposal is derived.

Conclusion

This framework provide the policy ground to develop a new unified continental system set with conditions to last. Many other frameworks could be designed around it and supporting its missions and different processes. The next steps suggest a simple collaboratio nacross african ministries of ecology to implement the surveillance and ecotoxicology measurement requirements and set up their financing and implement the political framework, all of which could be done within months and could lead to reduce dozens od gigatons of carbon emissions a year.